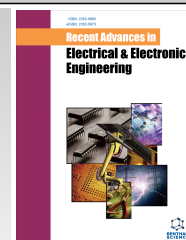


RESEARCH ARTICLE



Interpretability-driven Course Recommendation: A Random Forest Approach with Explainable AI



Tasleem Nizam¹, Sherin Zafar^{1,*}, Siddhartha Sankar Biswas¹ and Imran Hussain¹

¹Department of Computer Science and Engineering, Jamia Hamdard University, Delhi, India

Abstract: Introduction: In the current educational landscape, universities provide a vast array of diverse and obscure courses. However, the primary challenge students face is decision overload, as the abundance of options can make it difficult to choose courses that align with their interests, career aspirations, and strengths. The development and application of proper course suggestion systems might assist students in overcoming the challenges associated with choosing the right courses. However, these recommendation systems still require improvement to address explainability issues, security issues, and cold start. Our analysis reveals that there is limited research addressing how course recommendation systems can assist 12th grade students in selecting courses for higher education. Therefore, this study presents a novel personalized recommendation system, namely “Interpretability-Driven Course Recommendation: A Random Forest Approach with Explainable AI”, that chooses the top-3 courses for students based on their intermediate class grades/marks. This research provides accurate and interpretable recommendations using the Random Forest algorithm with Explainable AI that ensures transparency by explaining why a particular course is recommended.

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Methods: The system uses a Random Forest classifier with SHAP (SHapley Additive exPlanations) to forecast the top-3 courses with the greatest expected scores for appropriateness. The system uses SHAP (SHapley Additive exPlanations) values to integrate Explainable AI (XAI) to guarantee openness and trust. The proposed model solves the cold start problem and provides data security.

Results: In terms of precision (0.90), recall (0.92), and F1-score (0.92), Random Forest fared better than any other classifier. By combining predictions from several decision trees, its ensemble learning technique increases stability, decreases overfitting, and improves generalization across a broad range of input data patterns.

Discussion: A major drawback of conventional black-box machine learning models is their lack of transparency, which is addressed by the incorporation of SHAP into the course recommendation system. In this model, students get suggestions in addition to information on which topic scores had the biggest influence on their choice.

Conclusion: This study presents a practical and interpretable course recommendation system designed for 12th-grade students transitioning into higher education. Through the use of Explainable AI-enhanced Random Forest, the model provides precise, transparent, and customized recommendations. It addresses the main drawbacks of conventional systems, such as their lack of explainability and their cold start problems. The model's excellent precision, recall, and F1-score indicate its superior performance, which makes it a useful tool for student guidance and academic advice. For even more thorough recommendations, future research may consider combining long-term academic objectives, aptitude test results, and student interests.

Keywords: Random forest, cosine similarity, NLP, TF-IDF, precision, recall, F1-Measure, explainable AI (XAI).

* Address correspondence to this author at the Department of Computer Science and Engineering, Jamia Hamdard University, Delhi, India;
E-mail: sherin.zafar@jamiahamdard.ac.in

1. INTRODUCTION

With the wave of digitization, many schools and universities have developed online learning platforms and made their offline courses available online. These platforms allow students to choose courses that suit their needs. However, the abundance of course options requires students to spend significant time deciding on their preferred choices. Helping students make quick and effective course selections has become a pressing challenge for online education platforms [1, 2]. In recent years, recommendation technology has delivered remarkable results across various domains, such as online music platforms, e-commerce, and video content recommendations. Consequently, the application of recommendation technology to assist students in selecting suitable courses has gained increasing attention and popularity. As a result, the use of recommendation technology to assist students in selecting the right courses has become increasingly popular [3, 4].

In recent decades, significant advancements have been made in recommendation technologies. Course recommendations are commonly generated using traditional methods such as:

1.1. Content-based Recommendations

Using information from course descriptions and prior interactions with the platform, a content-based course recommendation engine generates course recommendations to students based on their unique interests and preferences [5]. After creating user and course profiles based on an analysis of course content attributes such as keywords, genres, and meta-data, these systems pair people with courses that are similar to them. In contrast to collaborative filtering, which depends on comparisons between users, content-based methods emphasize the intrinsic qualities of the courses. While effective in some scenarios, this approach is limited when user data is sparse [6].

1.2. Collaborative Filtering

A collaborative course recommendation system suggests courses that a user would find useful or interesting based on the preferences and actions of similar users [7]. To find trends and provide tailored recommendations, it examines information about how other users have interacted with courses, such as course ratings, enrollments, and completion rates. However, it profoundly suffers from the cold-start problem, where recommendations cannot be generated for new users or items due to a lack of historical interaction data [8].

1.3. Hybrid Approaches

Content-based filtering that examines course attributes, and collaborative filtering that examines user behavior and preferences, are combined in a hybrid course recommendation system to produce recommendations that are more varied and accurate [9]. In situations where there is a lack of data or where a user's preferences are not clearly defined, this strategy seeks to alleviate the shortcomings of individu-

al approaches by utilizing the advantages of both methods. But it still largely depends on prior user behavior, which is unavailable for newly enrolled students [10].

Machine learning techniques are frequently employed to improve the quality and accuracy of these recommendations [11]. For instance, P. Mishra *et al.* proposed a recommender system that combines machine learning algorithms with content-based filtering to tailor skills and courses based on user input [12]. Additionally, literature [13] advocated for the use of content-based filtering over context-based filtering when recommending colleges to students, concluding that content-based filtering delivers higher accuracy. A study [14] discussed systems for personalized recommendations. To assist online learners in efficiently navigating lecture videos, identifying key issues, and finding relevant articles to enhance their learning, the authors proposed an architecture for developing a machine learning-based Personalized Recommendation System (PRS). They found that the Random Forest method yields the most accurate predictions. Tran *et al.* [15] presented a course recommendation system utilizing pre-processing techniques and deep learning approaches with multi-layer perceptrons. Their model was proposed to advise students on course choices. Similarly, Zhen *et al.* [16] intended to increase the accuracy and personalization of recommendations by merging the Internet of Things (IoT) technology with deep learning (DL) to present a DL-based physical education course recommendation system.

Despite significant advancements, challenges remain in the field of course recommendation. For instance, one major issue is the cold-start problem, particularly for newly enrolled students. These students lack historical course selection data, relying only on basic details about their department [17]. Traditional approaches, such as content-based recommendation and collaborative filtering, struggle to accurately predict their interests under these circumstances. Some studies have attempted to incorporate more personalized features such as learning style, course difficulty, or peer feedback [18]. Others have explored recommendation systems for MOOCs (Massive Open Online Courses), focusing on behavioral data such as clickstream activity or quiz performance. While these efforts enhance personalization, they still overlook students who are new to the system or who have not interacted with online platforms extensively [19]. Additionally, very few studies take into account students' academic background, such as their intermediate (12th) grades, as a foundation for recommendations. This is a critical gap, especially for students entering college for the first time, where prior performance can be a strong indicator of academic preparedness and future course success.

Students consider multiple factors when selecting courses, including workload, the difficulty of maintaining a good GPA, and their interests [20]. Courses often come with a wide range of features, such as introductions, credits, and prerequisites. Once students gain a clear understanding of a course, they can typically decide whether to enroll in it or not [21]. Students rely on the course's diverse content to grasp its material. Therefore, effective course recommenda-

tions hinge on accurately predicting and interpreting the course content. However, a significant challenge in course recommendation systems lies in modelling the textual features of courses, as they are more complex to handle than their attribute-based features [22, 23]. Moreover, many systems use black-box machine learning models to optimize recommendation accuracy without offering any insight into why a specific course is recommended. This lack of explainability poses a challenge in educational settings, where transparency is crucial for trust and adoption by both students and educators.

Furthermore, while some recent research has explored the use of machine learning models like decision trees, k-NN, or neural networks in course recommendations, Random Forest models remain underexplored, particularly in combination with explainability frameworks like SHAP (SHapley Additive exPlanations) or LIME. These tools enable us to reveal the contribution of each feature to the recommendation decision, offering a layer of interpretability absent in many prior works.

At present, there are many research papers on course recommendation systems. But there are still some limitations of these papers that have not been resolved. These are –

- Issues such as cold start problem, privacy issues, data sparsity, data security problem *etc.*, these problems are still not resolved.
- The accuracy of existing papers is not so good. If the accuracy of any recommendation model is not satisfactory, then there is a chance that recommended items are not suitable for their users.
- Previous research papers have discussed courses that are either offered as a course in any college/ university, or they have conducted online exams and then based on students' marks, suggested courses. However, very few have directly addressed the marks of Intermediate pass students.
- There is a lack of transparency in suggesting courses in previous research papers, which also generates trust issues among students.
- Most existing systems focus on fixed rule-based approaches or black-box machine learning models, which lack transparency.
- Need for **Explainable AI (XAI) in educational recommendations** to ensure students and educators trust the system's suggestions.

Therefore, this study proposes a novel Interpretable Course Recommendation System: A Random Forest Approach with Explainable AI. This system generates recommendations using a Content Filtering Method that uses Cosine similarity and a random forest algorithm that utilizes the relationship between students and courses, based on different factors. This approach is particularly effective for new students with no prior course history, introducing a practical and underexplored approach. Also, with the help of explainable AI, the reason behind the generated courses is revealed. The following is a summary of this paper's contributions: A

novel solution to the cold-start problem in educational recommendation systems is proposed by leveraging students' intermediate (12th-grade) academic marks, bypassing the need for prior interaction or behavioural data.

- **Integration of Random Forest with content-based filtering** to improve recommendation accuracy using academic performance and course features, rather than relying on user interactions or survey-based preferences.
- **Use of Explainable AI (SHAP)** to make recommendations transparent and interpretable for students, addressing the lack of trust and transparency found in black-box models used in prior systems.
- **A practical recommendation model tailored to real-world student needs**, guiding learners toward courses aligned with their performance levels and skill development goals while promoting a balanced course enrolment.
- Empirical validation demonstrating that the proposed model outperforms traditional methods in both accuracy and user interpretability, especially in scenarios with no prior user data.

The rest of the article is structured as follows: Section 2 presents a concise literature review on course recommendation systems. Section 3 explores the recommendation methods along with problem formulation and its solution. The experimental setup is detailed in Section 4, followed by the results and an in-depth discussion in Section 5. At last, Section 6 summarizes the main conclusions and outlining possible avenues for further investigation.

2. LITERATURE SURVEY

The literature on different course recommendation systems is briefly analyzed in this section. Under this section, a quick overview of different works completed in the field of course-recommendation systems is given. The following studies demonstrate how alternative course recommendations may be made to students in order to save their time and effort while selecting courses based on their preferences or performance.

Researchers A.S. Kazi and A. Akhlaq [24] published their findings, identifying the factors influencing students' choices of careers. This study looked into the job choices made by 432 students from two universities in Lahore, Pakistan. The elements influencing their parental income, employment, and education were factors in decision-making. It also investigated how the respondents' choices to specialize in a certain field were influenced by their friends' and media sources' judgments.

Kang *et al.* [25] introduced a recommendation system for online broadcasting that was based on a tree model for personalized announcements. To reduce the overhead of preference predictions, real-time suggestions were generated by analyzing user preferences and used a hash map that combined the features of the tree.

In a work, V. Latha and S. Seethalakshmi [26] explored students' perspectives on their course selection experiences and their aspirations for professional advancement. About one hundred students were surveyed using a structured questionnaire to gain insight into the perspectives of the generation that would be changing the world in the future. The data were then subjected to statistical analysis.

A hybrid recommendation system based on user and content collaborative filtering was investigated by Li L. *et al.* [27]. This study investigated and analyzed recommendation systems that rely on user-collaborative filtering and content filtering. In this research, the conventional user-item rating matrix was replaced with a user-feature rating matrix that included item features and user ratings. The user set was subjected to K-means clustering, and suggestions were generated.

According to a study by I. Malhotra *et al.* [28], there is a method to help students choose the best optional courses for their areas of interest. With the use of the MF, or Matrix factorization approach, this strategy divided students into interest-based groups and then looked through previous student evaluation data in those domains to see the courses each given group member could enroll in.

J. W. Tzeng [29] discussed massively open online courses that are created based on advancements in instruction. They used mobile devices to encourage MOOC participation, developed a collection of practice questions for a MOOC recommendation system based on learners' online learning trajectories, and provided students with access to relevant practice questions *via* the chatbot on the social media platform LINE.

Tahir *et al.* presented the Deep Learning-based Course Recommendation System (DLCRS) that provides precise and learner-centered course recommendations by combining collaborative filtering techniques with a hybrid Sequential GRU+adam optimizer. For obtaining the learner's holistic perspective on the course's profiles and feedback, the suggested method integrates components for the extraction of learner features and course features, which is carried out using the (Embeddings from Language Models) ELMO word embedding technology [30].

Oladipo *et al.* presented a literature review in which authors used the Python programming language to overcome the issue of selecting elective courses by proposing an efficient system for recommending courses to undergraduate students based on previous students' grades. Students were categorized according to whether a given course would be beneficial for them, and they were presented with potential elective options derived from a logistic regression model, utilized alongside a comprehensive and sophisticated recommender system [31].

A. Jims *et al.* proposed a deep learning-based course recommendation system. Harris Hawks Optimization (HHO) and Deep Convolutional Neural Network (DCNN) were combined in their model (DCNN-CRS). Datasets from school data repositories were employed, features were retrieved using

HHO, and DCNN was used for classification. Their models were compared to CNN, LSTM, RESNET, and RNN models, and several characteristics were assessed, including accuracy, specificity, sensitivity, recall, and F1-score. By comparison of their model to others, it was shown that their model performed better [32].

Ogunde Adewale *et al.* [33] studied a collaborative filtering-based recommender system that uses undergraduate students' prior grades in relevant courses to suggest elective courses dynamically. This method used the k-nearest Neighbour technique to find hidden connections between the optional courses that are now offered and the related courses that students have already passed. The recommendation system was developed and tested using a dataset of actual student results. The summary of the above-reviewed papers is presented in the following Table 1.

3. METHODOLOGY

The architecture, problem formulation, and methods used for this research work are elaborated in this section. As shown in Fig. (1), our proposed system first takes the inputs from the user/student. If the student knows his grade/marks, then he will enter the marks/grades, and using the proposed recommendation algorithm, the suggestions will be generated. And if the user/ student is new (*i.e.*, cold start problem), then that user/student will be asked their preferences and based on their preferences, top-3 courses will be generated.

3.1. Problem Formulation

Given a set of students $S = \{s_1, s_2, \dots, s_n\}$ and a set of available courses $C = \{c_1, c_2, \dots, c_m\}$, the objective of the course recommendation system is to determine the most suitable course $Y_s \in C$ for each student s based on their academic performance. Each student s is characterized by a feature vector $X_s \in \mathbb{R}^k$, where $X_s = (x_1, x_2, \dots, x_k)$ represents the normalized average marks obtained in k different subjects. The recommendation system aims to learn a mapping function (Eq. 1):

$$F: X_s \rightarrow Y_s \quad (1)$$

This maximizes the probability of selecting the most appropriate course, formulated as (Eq. 2):

$$Y_s = \arg\max_{c \in C} P(c|X_s, \Theta) \quad (2)$$

where $P(c|X_s, \Theta)$ denotes the probability of selecting course c given student marks X_s and model parameters Θ . The system is designed to optimize a loss function $L(\Theta)$, expressed as (Eq. 3):

$$L(\theta) = \sum_{s \in S} l(Y_s, f(X_s; \theta)) \quad (3)$$

Where, $l(Y_s, Y_s^{\wedge})$ quantifies the discrepancy between the actual course Y_s and the predicted course $Y_s^{\wedge} = f(X_s; \Theta)$. Additionally, the system should ensure that recommendations adhere to academic eligibility constraints, where a course c is recommended to a student s only if the eligibility criteria are satisfied.

Table 1. Summary of literature survey.

Authors	Year	Title	Types of Algorithms used	Dataset Description	Key Findings	Strengths	Weakness
Asma Shahid Kazi <i>et al.</i> [24]	2017	Factors Affecting Students' Career Choice	Demographic variables Anova and t-test	432 students from two public sector universities in Lahore city	The findings demonstrate that the contribution of school counselors to the process of making well-informed career choices is minimal.	- Offers a thorough examination of the many elements influencing students' job decisions, such as media exposure, peer pressure, and parental influence. - Increases the study's depth by combining quantitative and qualitative methodologies.	- Restricts its direct application to recommendation system development by concentrating mostly on career choice rather than course suggestion. - Does not incorporate recommendation algorithms or technological solutions.
S. Kang <i>et al.</i> [25]	2020	Tree-Based Real-Time Advertisement Recommendation System in Online Broadcasting	Tree Model	Amazon Product Data	A tree model to predict a user's preferences in real-time. In	- The suggested tree-based recommendation system is appropriate for real-time applications in online broadcasting situations since it shows notable improvements in prediction latency. - When compared to conventional matrix-based techniques, the use of a hierarchical tree model enables effective preference prediction with less computing cost.	- Cold Start Problem - Lack of Explainability
V. Latha <i>et al.</i> [26]	2021	Analysis of Factors Influencing the Choice of Course Selection among Commerce Students in Higher Education	-	Survey	Analysed key points that affect commerce students	- The study offers insightful information about the variables, including peer pressure, employment opportunities, and personal interests, that affect commerce students' choice of courses. - Using qualitative approaches provides a more nuanced knowledge of how students make decisions, which can guide the creation of recommendation systems that are more focused on the needs of the students.	- The study's practical application in automated recommendation systems is limited because it does not include algorithms or technological solutions to solve the problems that were discovered. - The results may not be as broadly applicable to a larger student body if the study is limited to commerce students in a particular area.
Li L. <i>et al.</i> [27]	2021	Hybrid Algorithm Based on Content and Collaborative Filtering in Recommendation System Optimization and Simulation	Hybrid Filtering	MovieLens dataset and the scientific literature experimental dataset	Improved algorithm in this paper ensuring a better recommendation quality	- Combines content-based filtering with collaborative filtering, leveraging the advantages of both methods to improve recommendation accuracy. - Incorporates optimization strategies to enhance the performance of the recommendation system.	- Lacks information on the logic underlying suggestions, which could reduce user confidence. - The assessment ignores user pleasure and engagement levels in favor of concentrating mostly on accuracy indicators.

(Table 1) contd...

Malhotra et al. [28]	2022	A Course Recommendation System using Matrix Factorization and Domain-based Cluster Knowledge	Domain based-clustering and Matrix Factorization	-	Proposed system enhancing the overall learning experience for students.	- Clusters courses using domain knowledge, which could increase the recommendations' applicability. - Enhances recommendation personalization by revealing latent information through the use of matrix factorization algorithms.	- Does not specifically address the problem of new users. - Lacks the tools necessary to explain recommendations, which can be very important in learning environments.
J.W. Tzeng [29]	2023	Massive open online course recommendation system based on a reinforcement learning algorithm	Reinforcement Learning	RESTful APIs, using which a link has been created between the course website and database.	The actor-critic framework was applied to offer customized exercise recommendations within a MOOC platform.	- Uses an actor-critic framework to suggest exercises in an adaptable manner, gradually increasing personalization. - Increases accessibility and user engagement by delivering recommendations via a well-known messaging channel.	- Implementing the reinforcement learning approach might be challenging and computationally demanding. - Because the study was carried out on a particular MOOC platform, the results may not be as applicable in other settings.
Tahir et al. [30]	2024	Harnessing hybrid deep learning approach for personalized retrieval in e-learning	Course Recommendation System using Deep Learning	A reference dataset utilized within the Learning Management System (LMS) of a reputable public sector university	Enhancing e-learning platforms by implementing collaborative filtering to recommend course materials.	- Enhances tailored content retrieval in e-learning platforms by using deep learning algorithms. - Shows better performance metrics than conventional techniques, proving the hybrid approach's efficacy.	- The deep learning models that are employed serve as "black boxes," providing little information about how decisions are made. - The method might not work well in situations when there are new users and no previous interaction history.
Oladipo, I.D. et al. [31]	2021	An Improved Course Recommendation System Leveraging Historical Grade Data via Logistic Regression.	Logistic Regression	Department of Computer Science, The courses to be forecasted were solely electives within the University of Ilorin's department.	A model employed to assess students' suitability for specific courses and to suggest potential elective options for their consideration.	- Makes course recommendations based on students' prior academic success. - Makes use of logistic regression, which is simple to use and analyze.	- The accuracy of recommendations may be limited by logistic regression's inability to identify intricate, non-linear relationships in the data. - When scaling to larger datasets with more diverse student groups, the method can encounter difficulties.
A. Jims et al. [32]	2022	Course Recommendation System using the integration of Deep Convolutional Neural Network and Harris Hawks Optimization	Harris Hawks Optimization (HHO) and Deep Convolutional Neural Network (DCNN)	Datasets from school data repositories	Integrated model for recommending courses.	- Improves recommendation accuracy by combining deep convolutional neural networks with Harris Hawks Optimization. - Compared to baseline models, reports show increased recall and precision in course suggestions.	- Computational complexity is increased when deep learning and optimization methods are used together. - The intricacy of the model could make it more difficult to communicate advice to end users.
Ogunde et al., [33]	2023	Higher Education Recommendation Using KNN Algorithm	KNN	-	Model provided recommendation of colleges to students	- Makes recommendations for elective courses based on students' prior grades. - Predicts appropriate courses for students with an overall accuracy of 95.65%.	- Mainly depends on past academic data, which new students might not have access to. - As the size of the dataset increases, the K-NN approach may become extremely costly.

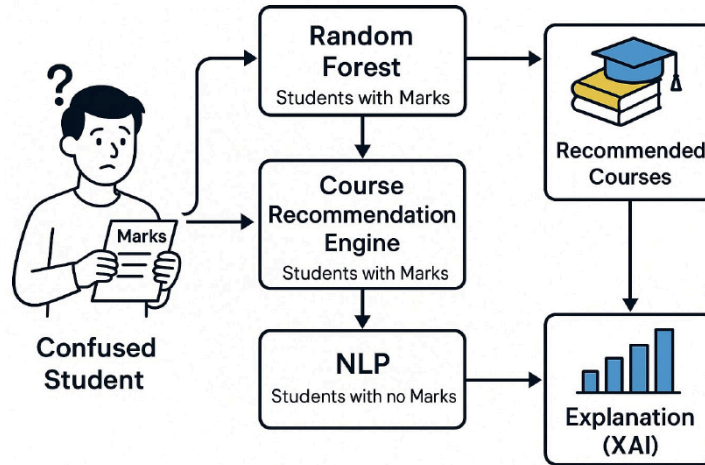


Fig. (1). Architecture of proposed recommendation system. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

Table 2. Description of course dataset's features.

Feature	Description of Feature
Name	Name of students
Subjects	Subjects (Physics, Chemistry, Maths) with their associated marks
Part Time	Checking student is enrolled part time or full time
Study Hours	Time studied by each student
Phone Number	Contact no of each student
Gender	Gender of each student
Grade	Grade secured by each student

3.2. Dataset Description

This paper collected the dataset from well-known public repositories, *i.e.*, Mendeley. This repository contains data from an anonymous school. Mendely provided a dataset for the courses with their ratings, which are used in this research. There are a total of 30 courses. These courses are engineering courses, such as B.Tech in computer science, B.Tech in mechanical engineering, B.Tech in petroleum engineering, B.Tech in electronics, *etc.*, that can be opted for by science students. The rating of these courses is done based on the market popularity or market demand of that particular course. Based on their ratings, these courses are assigned one more characteristic, *i.e.*, popular courses, intermediate courses, and low-enrolled courses. Table 2 is the description of the course dataset. The same repository provides the dataset of the students. This dataset contains the details of students in the 12th class. This student dataset contains the details of 9000 students. Every student has certain characteristics associated with them, such as their marks in subjects like physics, chemistry, and maths, their mobile no, their addresses, roll numbers, and student IDs. Table 3 presents a description of the student dataset. While dealing with these two datasets, some errors, like inconsistency, null values, *etc.*, were encountered, which were handled by appropriate desirable actions. So, this complete dataset contains two

profiles in itself, which are as follows:

3.2.1. Student Profile

It contains all the details of a student with their achieved marks. This database is selected after eliminating the undesirable characteristics from the original database. It includes students' names and all subjects (*i.e.*, Physics, Chemistry, Maths) with their marks.

3.2.2. Course Profile/Database

It contains all engineering courses. These courses have names, ratings, and course codes associated with them.

3.3. Used Algorithms

3.3.1. Random Forest

During the training stage, the random forest algorithm generates several Decision Trees. To evaluate a random assortment of features in every partition, a random sample of the training data and features from the dataset is employed to create every tree. By fostering diversity among the individual trees, this randomness reduces the likelihood of overfitting and enhances the overall accuracy of predictions [34]. The generated trees learn decision rules (*e.g.*, if Math > 80

Table 3. Description of student dataset's features.

Feature	Description of Feature
Course Name	Name of different courses
Ratings	Course's Ratings
Course Code	Specific code name associated with each course

and $\text{course_rating} > 4.0 \rightarrow \text{recommend}$). For a new student, the trained model takes their marks and course data as input. Then each tree makes a recommendation (vote). The algorithm synthesizes the outputs from all trees for prediction purposes, using averaging for regression tasks and voting for classification tasks, as depicted in Fig. (2). An illustration of dependable and precise results is demonstrated by this joint decision-making approach, which benefits from the perspectives of various trees. Random forests are often employed for classification and regression purposes, as they effectively handle complex datasets, mitigate overfitting, and yield accurate forecasts in a wide range of contexts [35].

3.3.2. Cosine Similarity

Regardless of the magnitude difference between two vectors, cosine similarity is a metric used to assess how similar they are. Because of its broad application across multiple domains, such as information retrieval, text mining, and recommendation systems, it is a popular tool for comparing the similarity of objects like user profiles, papers, and item descriptions [36]. Cosine similarity is calculated by finding the cosine of the angle between two non-zero vectors in an inner product space, with values ranging from -1 to 1 (Eq. 4), where:

$$\text{Cosine similarity } (A, B) = \frac{A \cdot B}{(\|A\| \|B\|)} \quad (4)$$

Here,

- The dot product of vectors **A** and **B** is represented as $\mathbf{A} \cdot \mathbf{B}$.
- The magnitude of vector **A** is denoted as $\|\mathbf{A}\|$.
- The magnitude of vector **B** is denoted as $\|\mathbf{B}\|$.

3.3.3. Natural Language Processing

A machine learning technique called natural language processing (NLP) enables computers to comprehend, modify, and interpret languages used by humans [37]. In this manuscript, we have used "Text Vectorization and similarity-based NLP" to ask about the interests of students. Using techniques like "Word Embeddings" or "Bag-of-Words", which map words to numerical vectors so that similar words have similar vector representations, text vectorization in natural language processing (NLP) is the process of turning raw text into machine-understandable numerical representations (vectors) that allow for analysis by comparing the similarity between these vectors, essentially measuring how close two pieces of text are in meaning. This makes it possible to perform tasks like sentiment analysis or document classification based on text similarities [38].

3.3.4. Explainability in Course Recommendation using SHAP

Explainability in course recommendation systems enhances transparency, trust, and fairness by providing clear justifications for why a particular course is recommended to a student. Instead of a "black-box" approach, explainability allows students, educators, and administrators to understand and validate the decision-making process of AI-based recommendations. SHAP (SHapley Additive exPlanations) is a popular Explainable AI (XAI) technique used to interpret the decisions of machine learning models, including course recommendation systems [39]. SHAP assigns a contribution value to each feature, explaining how much each factor (e.g., student marks, subject scores, past performance) influences the recommendation of a particular course. SHAP values are derived using Shapley values from cooperative game theory. The formula for computing SHAP values is (Eq. 5):

$$\phi_i = \sum_{S \in N \setminus i} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (5)$$

Where, ϕ_i = SHAP value for feature i

S = A subset of features excluding i



Fig. (2). Architecture of the random forest algorithm.

- The value of 1 denotes that the vectors are equivalent (or point in the same direction).
- The value of 0 denotes that the vectors are orthogonal (no similarity).
- The value of -1 denotes that the vectors are diametrically opposed.

N = Total number of features in the model.

$f(S)$ = The model's output (course recommendation probability) using feature set S

$f(S \cup \{i\})$ = The model's output when feature i is added.

The fraction represents the weighted average of all possible feature contributions [40].

3.4. Proposed Solution

The course recommendation is a system that selects courses from a large set of courses. These courses are selected by creating a match between the student's marks/grades and the rating of courses. The students having higher average marks/higher grades get those courses which are of higher ratings, intermediate average-marked students get intermediate rating courses, and lower average-marked students get lower rating courses. For instance, if student A has secured an 80% average mark, then he will be suggested the top-3 courses from the course-dataset.

There is a set of attributes in the student dataset that influences course recommendation. Let $S = \{s_1, s_2, s_3, \dots, s_n\}$ be a set of students. Each student S_i has some attributes. Let $X = \{x_1, x_2, x_3, \dots, x_n\}$ be a set of attributes that are associated with the student-dataset, like the student's name, mobile number, address, marks of each subject, *etc.* Among these attributes, some are used while making the course recommendation. For instance, attributes like Maths, Physics, and Chemistry are used to calculate the average marks of each student, which are then used for course-suggestion purposes.

There is a set of attributes also in the course dataset, which are used for recommendation. Let $C = \{c_1, c_2, c_3, \dots, c_n\}$ be a set of courses that are available in the course dataset. Each course has a level associated with it. For instance, based on the rating of courses, each course's popularity level is associated. For example, the courses having higher ratings are considered "Outstanding courses", having moderate ratings are considered "Average courses", and having lower ratings are called "Below average courses".

In this manuscript, the random forest algorithm is used for course recommendation. Under the Random-forest algorithm, each tree learns from a bootstrapped sample of the training dataset and uses a subset of features to make predictions. The goal is to recommend a suitable course (C) from a set of possible courses ($\{C_1, C_2, \dots, C_k\}$) for students based on their obtained average marks (M) and other relevant features (X). So, the inputs are:

- M : average marks of the student (primary feature).
- $X = [x_1, x_2, \dots, x_p]$ additional features (*e.g.*, marks of each subject).
- Y : target variable representing the course to recommend (C).

And the output will be

- Predicted course Y' for the student.

Initially, the feature space/feature vector will be defined. It is defined as (Eq. 6)-

$$Z = [M, x_1, x_2, \dots, x_p] \quad (6)$$

where:

- M : Average marks (numerical).
- $x_1, x_2, \dots, x_p$: Other features such as marks of each subject.

After defining the feature space, the training of the data is done there. Proper training of data leads to the right suggestions for courses. This training is done by considering the average marks of each student and the ratings of each course. The input data/training dataset is formulated as (Eq. 7)-

$$\{(Z_i, Y_i)\}_{i=1}^N \quad (7)$$

where,

- Z_i is the feature vector
- Y_i is the actual course taken by student i .

After giving the input dataset, each decision tree h_i is trained on a random subset of $\{(Z_i, Y_i)\}$ and splits features to minimize impurity (*e.g.*, Gini index or entropy for classification) (Eq. 8):

$$\text{Impurity}(S) = \sum_{j=1}^k p_j (1 - p_j), \quad (8)$$

where p_j is the proportion of samples in class j .

Given a new student with feature vector $Z_{\text{new}} = [M_{\text{new}}, x_1, x_2, \dots, x_p]$:

- Each tree h_i predicts a course:

$$h_i(Z_{\text{new}}) \in \{C_1, C_2, \dots, C_k\}$$

- Then the Random Forest Algorithm aggregates the prediction:

$$Y' = \text{mode}(\{h_1(Z_{\text{new}}), h_2(Z_{\text{new}}), \dots, h_T(Z_{\text{new}})\})$$

Finally, the predicted course will be generated (Eq. 9):

$$Y' = \frac{1}{T} \sum_{t=1}^T h_t(Z_{\text{new}}) \quad (9)$$

Since this manuscript solves the cold-start problem, the cold-start problem is solved using Text Vectorization and similarity-based NLP. For this purpose, the Term Frequency-Inverse Document Frequency (TF-IDF) is used. Initially, students without any mark details will be asked to provide their preferences for courses. Then their textual data and course dataset both will be converted into a bag-of-words so that numerical computation on these textual data can be done. To enable numerical computation, the textual data must be converted into vectors, for this, TF-IDF method is

used.

TF-IDF is a method used to assess the significance of a word in relation to a specific text within a collection of documents. The occurrence frequency of a word in the corpus (dataset) enhances its importance, which correlates with the number of times it appears in the text. TF-IDF can be formulated as (Eq. 10-12):

$$TF - IDF(t, d) = TF(t, d) \times IDF(t) \quad (10)$$

where,

$$TF(t, d) = \frac{\text{Frequency of term } t \text{ in the document } d}{\text{Total terms in } d} \quad (11)$$

$$IDF(t) = \log \frac{\text{Total number of documents}}{\text{Number of documents containing } t} \quad (12)$$

Once both student preferences and course descriptions are vectorized, the cosine-similarity measure is applied. Once the cosine similarity is calculated, courses are ranked based on similarity scores in descending order. After ranking the courses in descending order, the top 3 courses will be recommended to the student.

After suggesting the top-3 courses, the SHAP method is used to explain how the input features impact course recommendations.

4. EXPERIMENTAL SETUP

Under this section, we discuss our baselines, steps of the algorithm (using Random Forest), and evaluation metrics.

Baselines: We compare our model against the following methods-

4.1. Logistic Regression

For binary classification, logistic regression uses the sigmoid function, which takes independent variables as input and produces an anticipated value between 0 and 1. For instance, Class 0 and Class 1 are two instances of our classes. An input is classified as Class 1 if its logistic function value is greater than the cutoff value of 0.4; otherwise, it is classified as Class 0. It is called regression because it is a continuation of linear regression [41]. But it is mostly used for classification problems. Some authors [42] use Logistic Regression for suggesting courses to the students that take into account both the instructors' lesson plans and the individual knowledge traits of the students. The authors found that their model produces 0.76 and 0.84 accuracy. Under this research, Logistic Regression is implemented using the Scikit-learn library in Python. The configurations of the Logistic Regression algorithm are described as follows:

- **Solver (solver='lbfgs'):** The model employs the Limited-memory Broyden-Fletcher-Goldfarb-Shanno (LBFGS) algorithm, a quasi-Newton optimization technique well-suited for small to medium-sized datasets. It is efficient for convex loss functions and supports multinomial loss for multiclass problems.
- **Regularization Penalty (penalty='l2'):** L2 regularization,

also known as Ridge regularization, is applied to prevent overfitting by penalizing large coefficients. This ensures that the model generalizes better to unseen data.

- **Inverse Regularization Strength (C=1.0):** The regularization strength is controlled by the parameter C, which is the inverse of the regularization coefficient (*i.e.*, smaller values imply stronger regularization). The default value of 1.0 represents a balanced trade-off between bias and variance.
- **Maximum Iterations (max_iter=100):** The optimization algorithm is allowed to run for a maximum of 100 iterations to converge to a solution. This value is sufficient for most well-conditioned datasets.

4.2. KNN

If there is a data/database that contains missing values, in that case, KNN can be used. KNN is a straightforward supervised machine learning algorithm that is extensively used in regression or categorization tasks. Its basic rule is that while collecting data, those data points that are closest to a distinct point are considered "identical" observations; thereby, one can group the unanticipated data points correspondingly to the values of the nearest existing data points. Or, to put it another way, it is a non-parametric approach that assumes that items that are similar to one another are also likely to be close in proximity, either because they share the same class or possess similar attributes [43]. Researchers [44] used the KNN algorithm to suggest the nearest colleges to the students for their higher education. Using Scikit-learn in its default configuration, the K-Nearest Neighbors classifier was implemented in this research. The following is a description of the pertinent parameters:

- **Number of Neighbors (n_neighbors=5):** The model considers the 5 closest data points in the feature space to determine the class of a given input. This value balances bias and variance in the absence of hyperparameter tuning.
- **Distance Metric (metric='minkowski', p=2):** The distance between data points is calculated using the Minkowski distance, with p=2, which corresponds to the Euclidean distance. This metric is commonly used due to its geometric interpretability.
- **Weighting Function (weights='uniform'):** All neighbors contribute equally to the classification decision. That is, each of the k nearest neighbors has an equal vote in determining the output class.
- **Algorithm (algorithm='auto'):** The implementation automatically selects the most appropriate algorithm for computing nearest neighbors based on the training data size and structure, choosing between 'ball_tree', 'kd_tree', and 'brute'.

4.3. Decision Tree

One of the most practical supervised learning algorithms available is the Decision Tree (DT). Decision Trees are

Table 4. Comparison of proposed methodology with existing works.

Parameter/s	Course Recommendation System using Machine Learning [47]	An Improved Course Recommendation System Based on Historical Grade Data Using Logistic Regression [42]	Higher Education Recommendation Using KNN Algorithm [44]	DORIS [50]	Our Proposed Research work
Used Methodology	Decision Tree	Logistic Regression	KNN Algorithm	Deep Learning model	Random Forest with Explainable AI
Feature Interaction	Limited Interactions	High level feature interaction	High Level of feature Interaction	High order of feature interaction	Limited interaction with individual classifiers
Performance Metrics	×	Accuracy, MSE	×	Surpasses alternative techniques in terms of recommendation quality.	Precision, Recall, F1-measure
Experimental Results	Better than Traditional Methods	Promising results	Results are fine	Shown better performance than baselines used under DORIS	Higher than all baselines (i.e., KNN, Logistic Regression and Decision Tree) methods
Transparency in results	×	×	×	×	√
Solution for cold start problem	×	×	×	×	√
Data Security	×	×	×	×	√

machine learning methods. It gradually splits large sets of data into tiny groups of data based on a descriptive feature, until they reach sets which are tiny enough to be labelled. They attempt to label new data according to the information they have about the tagged data, which they require. Decision Tree algorithms are ideal for solving problems related to regression (where the system estimates values, such as a building price) and classification (where the system classifies data into classes, such as whether an email is fraudulent or not). The fact that Decision Trees have numerous real-world applications is what makes them so important [45]. Since it is one of the most widely used ML algorithms, it is utilized for various purposes across numerous sectors [46]. Some authors have [47] used the Decision Tree algorithm for a course recommendation system

to offer practical decision-making pathways for lifelong learning tailored to individual professional needs. Additionally, Scikit-learn's default configuration is used to create the Decision Tree classifier for our research. The following defines the important parameters:

- **Splitting Criterion (criterion = 'gini'):** The model uses the Gini Impurity measure to evaluate the quality of a split. It favors balanced splits that isolate classes effectively, contributing to lower classification error.
- **Tree Depth (max_depth = None):** There is no preset limit on the maximum depth of the tree. This allows the tree to grow until all leaves are pure or until all samples are exhausted, which may lead to overfitting if not controlled through pruning or other regularization strategies.
- **Minimum Samples to Split (min_samples_split = 2):** A node must have at least two samples to be con-

sidered for splitting. This is the smallest value allowed, enabling the tree to expand rapidly unless constrained.

- **Minimum Samples per Leaf (min_samples_leaf = 1):** A leaf node can contain as few as one training sample, allowing for maximum granularity in classification at the risk of overfitting.
- **Splitter Strategy (splitter = 'best'):** The best split is chosen based on the highest information gain for each node, ensuring locally optimal decisions during tree growth.

4.4. Deep Learning

Deep Learning is a branch of machine learning that focuses on artificial neural network algorithms that are modeled after the composition and operations of the human brain. Without the need for human feature engineering, it allows computers to automatically recognize intricate patterns and draw conclusions from data [48]. Deep learning methods use several layers of interconnected neurons to automatically learn features and representations from raw data, in contrast to typical machine learning models (such as Decision Trees and Logistic Regression), which may have trouble with extremely nonlinear or high-dimensional data [49]. Ma *et al.* [50] presented research that chooses the best courses for students based on their interests, basic facts, and course specifics. For the proposed research work, the PyTorch framework was used to create a deep learning-based recommendation model in order to provide a more thorough performance benchmark. The model learns non-linear correlations between course parameters (like ratings) and student academic data (like marks) by utilizing a fully linked Multi-Layer Perceptron (MLP) architecture. Two hidden layers, each with 64 and 32 neurons, respectively, and ReLU (Recti-

fied Linear Unit) activation functions, follow the input layer, which is based on the number of features. To lessen overfitting, a Dropout layer is inserted after the first hidden layer, with a default probability of 0.2. One sigmoid-activated neuron in the output layer is appropriate for binary classification, which determines whether a path should be suggested (1) or not (0) (Table 4).

4.5. Applied Algorithm

Our model, known as “Interpretability-Driven Course Recommendation: A Random Forest Approach with Explainable AI”, works on the Random Forest Algorithm. The working steps of this model are defined as follows: Algorithm 1:

Input: Dataset, End User Query

Output: Recommended Courses

- (1) Import student data D.
- (2) Prepare Features and Target. Let X = features (student average marks) and Y = Target Variable(recommended Courses).
- (3) If the student has their marks, then go to step VIII, else go to step IV.
- (4) Ask the student about their interest.
- (5) Preprocess text data. Use TF-IDF (Eq. 13):

$$TF - IDF(t, d) = TF(t, d) \cdot \log \frac{N}{DF(t)} \quad (13)$$

where $TF(t, d)$ = Term frequency of term t in document d .

$DF(t)$: Number of documents containing term t .

N : Total number of documents.

- (1) Combine numerical and textual features into a single feature matrix X (Eq. 14):

$$X = [X_{numerical} | X_{textual}] \quad (14)$$

- (1) Extract key features from the combined dataset.
- (2) Divide D into training and testing sets.
- (3) Initialize the Random Forest Model, which is an ensemble of N decision trees. For each tree:
- (4) Select a random subset of training data (with replacement) for building the tree (bootstrap sampling).

Select a random subset of features at each split to reduce overfitting.

- (1) Repeat steps from a to c until max depth is reached

For each decision tree at each split (node), calculate the Gini Impurity to determine the best feature and threshold to split the data (Eq. 15).

$$Gini\ Impurity\ (G) = 1 - \sum_{i=1}^c p_i^2 \quad (15)$$

where C is the number of classes and p_i is the proportion of class i in the node.

For each feature and threshold, calculate (Eq. 16):

$$\text{Reduction in Gini Impurity } \Delta G = G_{parent} - \left(\frac{N_{left}}{N} G_{left} + \frac{N_{right}}{N} G_{right} \right) \quad (16)$$

Choose the feature and threshold that minimize G .

- (1) After training N decision trees, the Random Forest aggregates the predictions using majority voting (Eq. 17):

$$Y' = Mode(Y_{tree1}, Y_{tree2}, \dots, Y_{treeN}) \quad (17)$$

- (1) For a new student, pass their features X_{new} through all N trees in the forest.
- (2) Aggregate the predictions to recommend the most likely course (Eq. 18):

$$Y'_{new} = Mode(Y_{tree1}(X_{new}), Y_{tree2}(X_{new}), \dots, Y_{treeN}(X_{new})) \quad (18)$$

- (1) Recommended courses.
- (2) Use the SHAP formula for the explainability of the course suggestion.

4.6. Evaluation Metrics

In this paper, some evaluation metrics are used for verification purposes. These metrics help in determining how much the recommendation provided by any system is correct. These metrics are-

- Precision: The proportion of the real positive results to the entire number of positive results, the model anticipated is provided by the precision metric (Eq. 19). The question “How much of our optimistic forecasts came correct?” is answered by it [51].

$$Precision = \frac{\text{true positives}}{\text{true positives} + \text{false positives}} \quad (19)$$

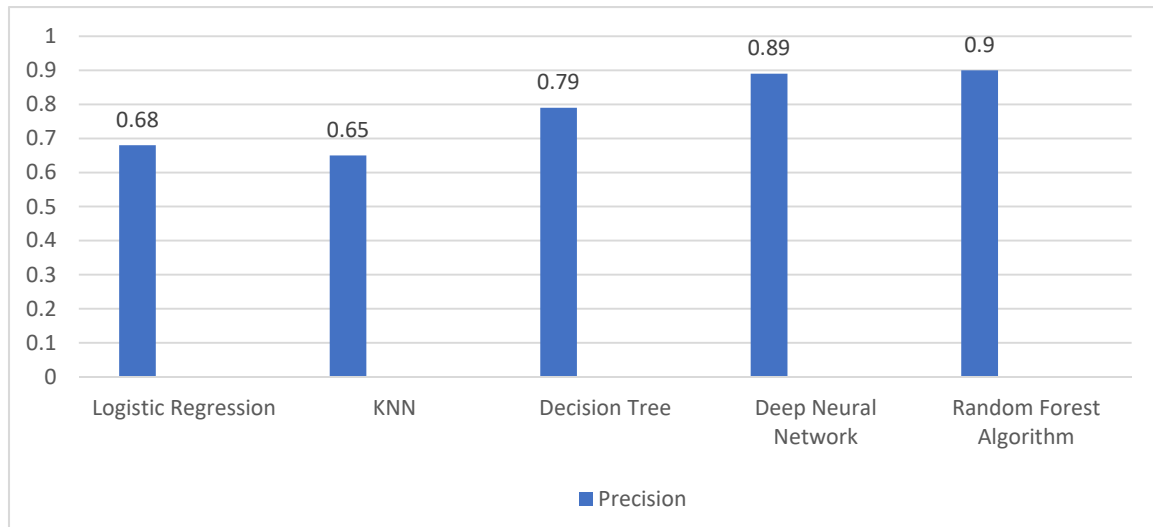
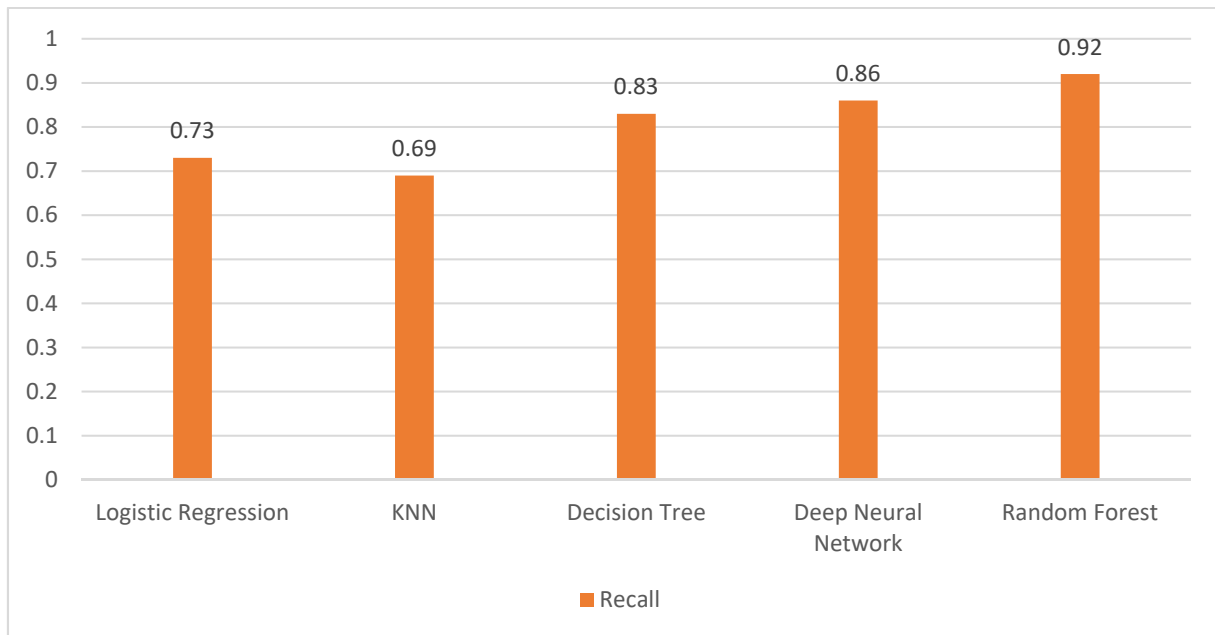
- Recall: The model's recall quantifies its capacity to recognize each positive. Recall is also referred to as the true positive rate (Eq. 20). It provides the answer to the query: “How much of each of the data points that should have been predicted as true did we correctly predict as true [52]”?

$$Recall = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \quad (20)$$

- F1-measure: A single statistic known as the F1-measure combines recall and precision. It is found that the F1-measure is a useful metric to assess how effectively any model makes the trade-off between recall and precision (Eq. 21). An essential component of the score is that the F1-measure is 0 if either recall or precision decreases to zero. It discourages exceedingly small values of any element as a result [53].

Table 5. Results on proposed and baseline algorithms.

Algorithm Name	Precision	Recall	F1-measure
Logistic Regression	0.68	0.73	0.70
KNN	0.65	0.69	0.67
Decision Tree	0.79	0.83	0.81
Deep Neural Network	0.89	0.86	0.87
Random Forest Algorithm	0.90	0.92	0.91

**Fig. (3).** Chart of precision values of logistic regression, KNN, decision tree, deep neural network, random forest. (A higher resolution / colour version of this figure is available in the electronic copy of the article).**Fig. (4).** Chart of recall values of logistic regression, KNN, decision tree, deep neural network, random forest. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

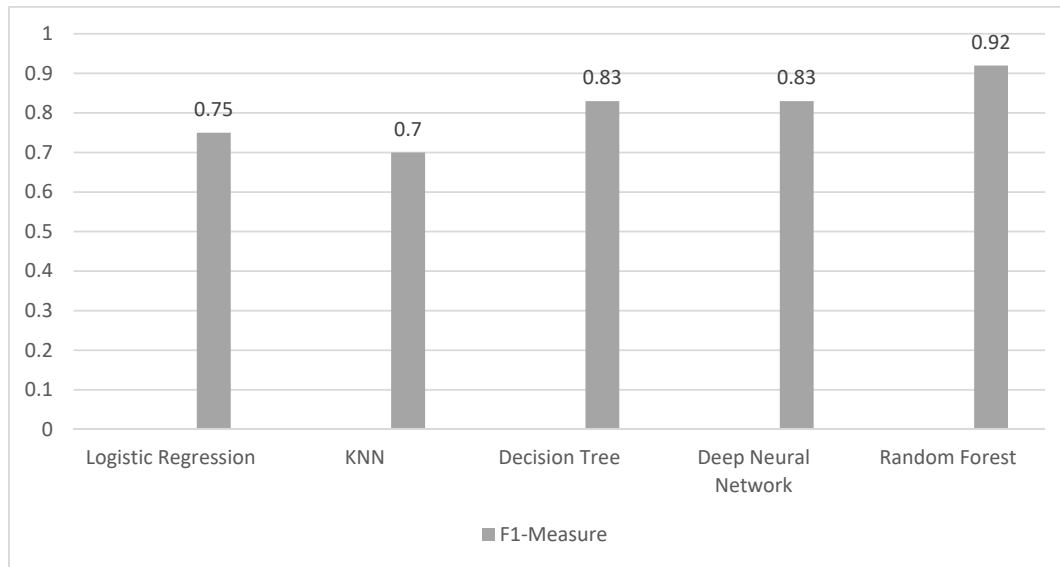


Fig. (5). Chart of F1-measure values of logistic regression, KNN, decision tree, deep neural network, random forest. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

$$F1 \text{ measure} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (21)$$

5. RESULTS AND DISCUSSION

After applying the baseline algorithms (*i.e.*, KNN, Logistic Regression, Decision Tree, and Deep Neural Network) and the Random Forest algorithm on the presented datasets, we found the following results (shown in Table 5) in terms of accuracy. By observing Table 5 below, it is concluded that our proposed methodology (Random Forest Algorithm) performed better than the baselines.

The chart for the comparison of all these algorithms, *i.e.*, Logistic Regression, KNN, Decision Tree, Deep Neural Network, and Random Forest Algorithm, is given in Figs. (3-5) respectively.

From the above Table 5 and Figs. (3-5), it can be observed that KNN gave the worst performance among the three machine learning algorithms. The reason behind it is that the relationship between marks and courses is non-linear, which results in the degradation of the performance of KNN. Additionally, KNN is not a suitable algorithm if the dataset size is large. If there are anticipated significant correlations between grades and course preferences and the dataset is short, KNN is a smart choice.

Logistic Regression improves performance over KNN, as shown in Table 5 and Fig (3-5). But its performance is not better than Decision Tree. This is so because Logistic Regression performs well in binary classification. Additionally, it assumes a linear relationship between courses and marks, even though the relationship between courses and marks is non-linear, and there are many courses for which Logistic Regression cannot perform well.

Decision Tree works better than both KNN and Logistic Regression, as it is shown in Table 5 and Figs (3-5). But it could not beat the Random Forest algorithm because Decision trees tend to overfit, especially if the tree depth is not limited. This can lead to poor generalization on unseen data. Additionally, A single tree is more sensitive to noise and outliers in the dataset, leading to lower accuracy compared to Random Forest.

Tree-based models manage the majority of the numerical and categorical input characteristics (such as scores and gender) that are used in the course recommendation system. Unless feature interactions are intricate and high-dimensional, which is not the case in this research, deep learning does not yield noticeable benefits. This kind of structured data is better suited for Random Forest since it automatically records feature interactions without the need for feature scaling or embedding layers.

In our research, Random Forest performed better in terms of accuracy than Deep Neural Networks. This is because the dataset (9000 records) is organized and relatively small, making it more appropriate for tree-based ensemble algorithms. Conversely, DNNs are more likely to overfit in low-data regimes, need extensive hyperparameter adjustment, and require huge amounts of data.

The Random Forest Algorithm outperformed every baseline algorithm, the reason behind it is that Random Forest is an ensemble learning method that builds multiple decision trees and aggregates their predictions (majority voting for classification or averaging for regression). Compared to a single decision tree, this reduces the chance of overfitting. Furthermore, by merging predictions from several trees, Random Forest generalizes better and is less sensitive to small

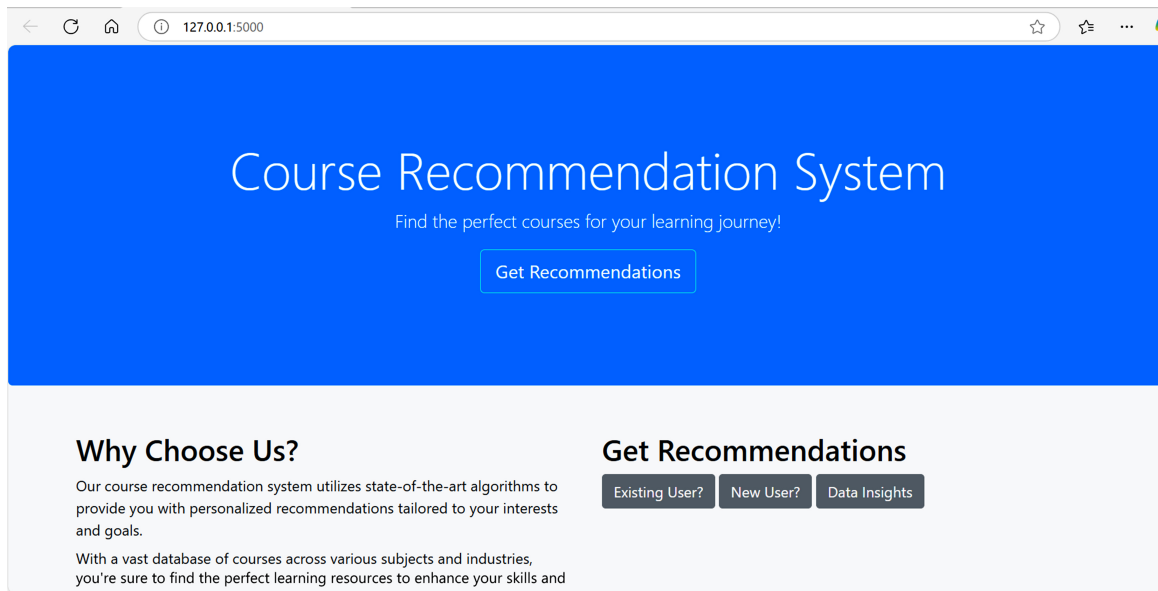


Fig. (6). Homepage of proposed system. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

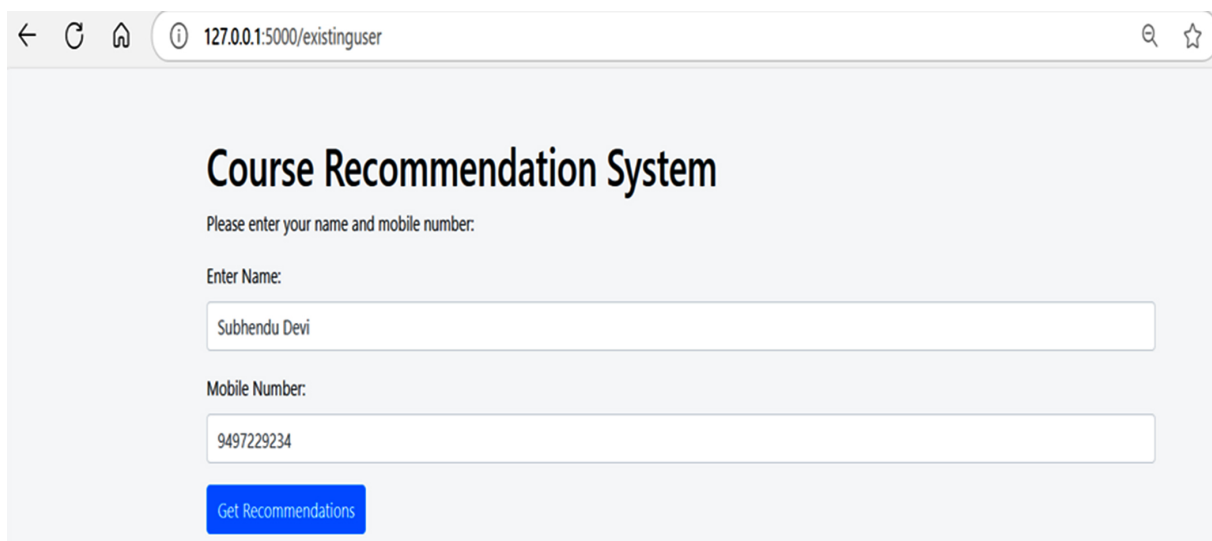


Fig. (7). Inputting details of an existing student. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

changes in the dataset. Snapshots of the prediction of courses under this research are shown in the following Figs. (6-10).

5.1. Explainability using SHAP

To interpret the model's decision-making process, SHAP (Shapley Additive Explanations) summary plot analysis

and SHAP dependence plot analysis are used to analyze feature importance and individual feature contributions.

5.1.1. SHAP Summary Plot Analysis

The SHAP summary plot (Fig. 11) highlights the most influential features in the course recommendation process. The colour represents the feature value (red = high, blue = low). Features are ranked from the most important (top) to

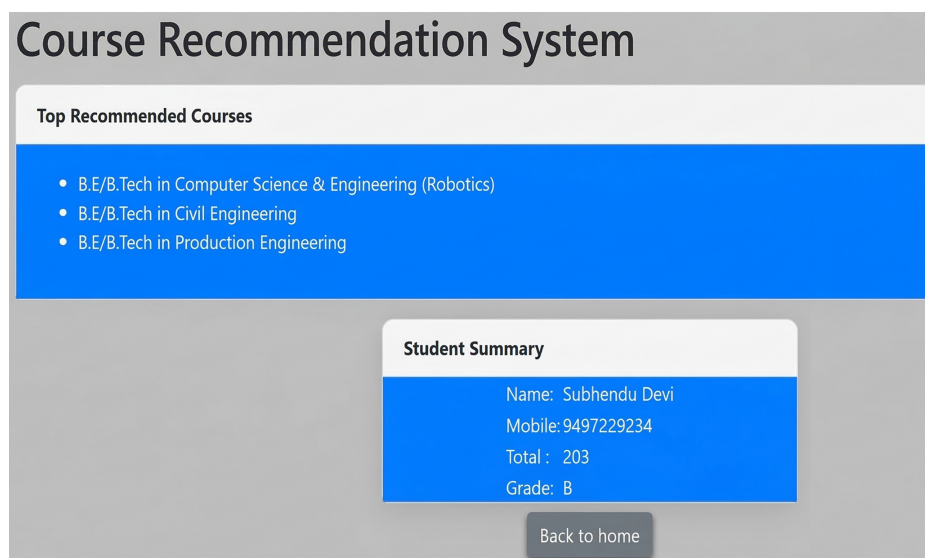


Fig. (8). Suggestion of Top-3 courses. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

The screenshot shows the registration form for the 'Course Recommendation System'. It includes input fields for 'Name' (filled with 'Sanjeev Rai') and 'Mobile Number' (filled with '9988776655'). Below these, there are radio buttons for 'Marks', 'Grade', and 'Preference', with 'Preference' selected. A 'Preference' dropdown menu is shown with 'Networking' selected. A blue 'Get Recommendations' button is positioned at the bottom left of the form area.

Fig. (9). Details entering for new user. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

the least important (bottom). The x-axis shows SHAP values, which indicate how much a feature contributes to making the prediction higher or lower. It can be shown as, the Average Marks feature emerged as the most critical determinant, significantly impacting the model's predictions. Other relevant features include Mathematics Marks, Chemistry Marks and Physics Marks, indicating that students with higher scores in STEM (Science, Technology, Engineering, and Mathematics; in context to this research paper, it is Mathematics, Physics, and Chemistry) subjects are more likely to be recommended for technical courses. The summary plot also reveals a positive correlation between higher subject marks and the likelihood of being recommended for engineering disciplines. In contrast, students with lower STEM

scores are more frequently assigned to non-technical courses, showcasing the model's adherence to academic strengths.

5.1.2. SHAP Dependence Plot Analysis

We used SHAP (SHapley Additive exPlanations) dependent plots to improve the interpretability of the course recommendation model. These plots show how each input feature affects the model's predictions for various classes. In particular, we concentrated on the academic disciplines of mathematics, physics, and chemistry, which are crucial factors in engineering admissions. The TreeExplainer for our trained Random Forest model was used to calculate SHAP values. Because our problem involved multiple classes, SHAP

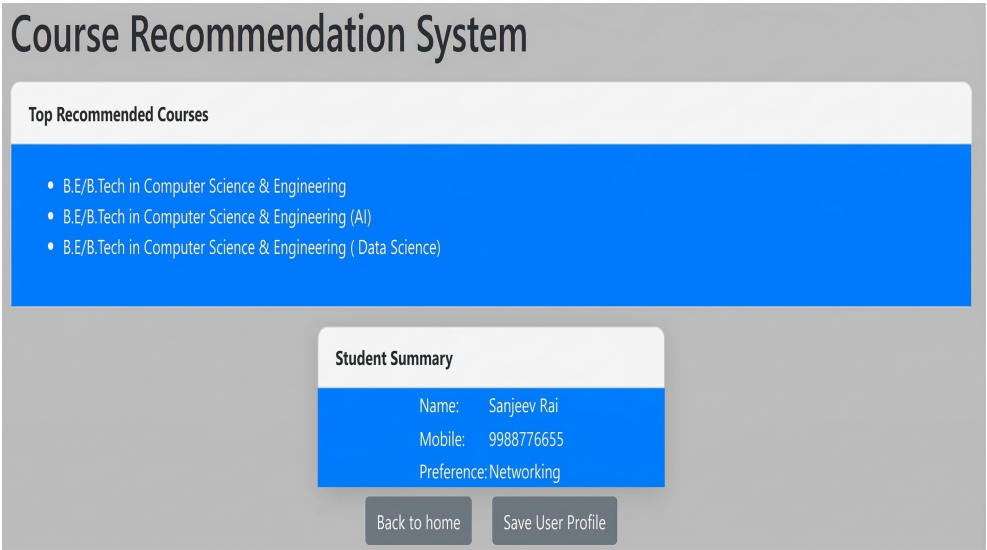


Fig. (10). Recommended courses to new users. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

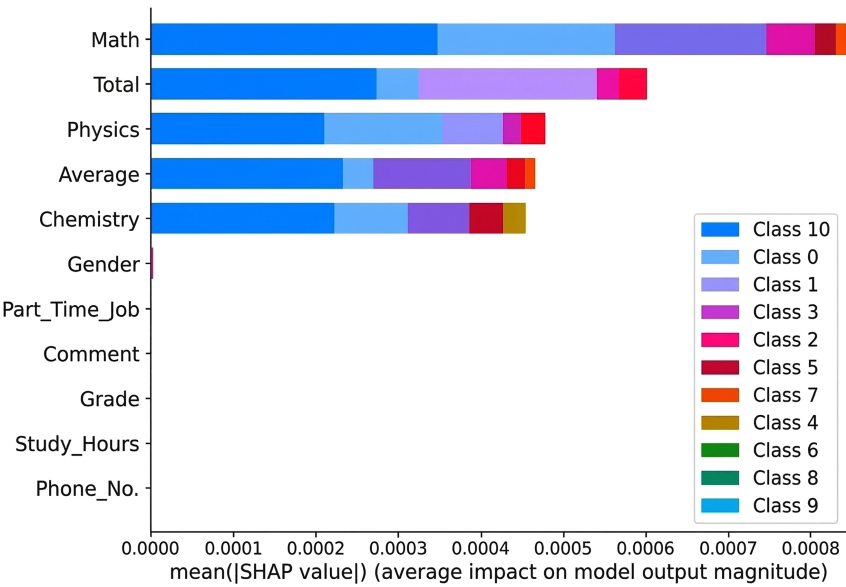


Fig. (11). A SHAP summary plot analysis of features. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

values were determined independently for each target class, which stands for distinct engineering courses. We created dependent graphs for every feature in every class-specific output for ease of interpretation and visual clarity. The colour represents the feature value (red = high, blue = low). The X-axis shows the actual value of the input feature that were analyzed. For example, when plotting the feature for “Physics”, then the x-axis will show the student’s real score, which ranges from 0-100. The Y-axis displays each data point’s feature’s SHAP value. For a given instance, a SHAP value indicates the feature’s marginal contribution to the final model

prediction. Following is the description of how each subject is affected in each class:

• **Physics**

Higher Physics grades are strongly correlated with the recommendation probability for engineering disciplines that heavily rely on physics, such as mechanical and electrical engineering, according to the SHAP dependence plots (Fig. 12). Pupils who score higher than a particular criterion have significantly higher SHAP values, which suggests that they

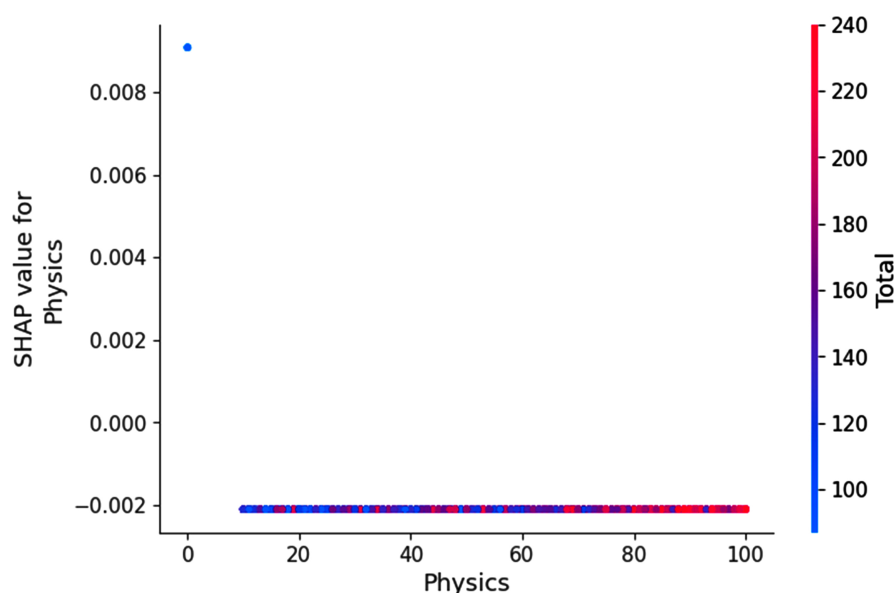


Fig. (12). Shap dependence value for physics. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

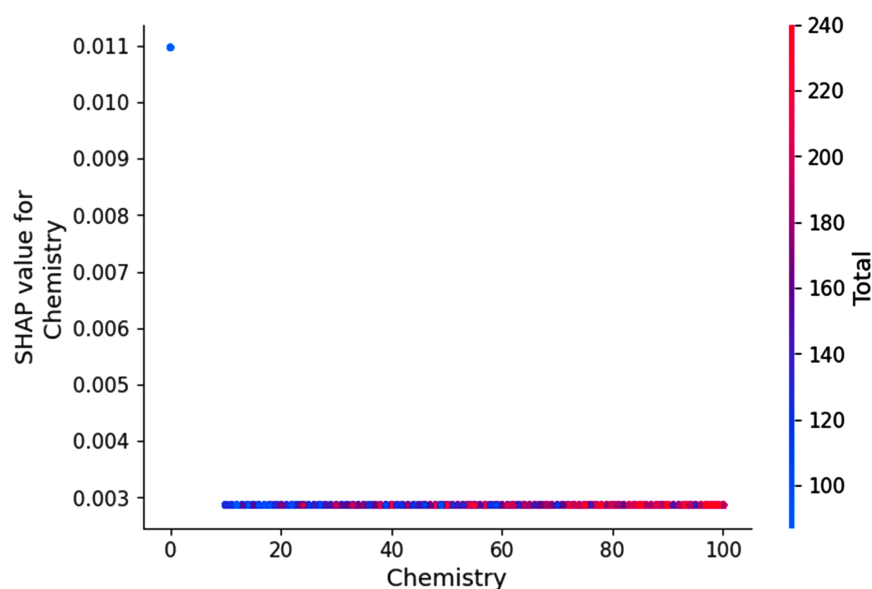


Fig. (13). Shap dependence value for chemistry. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

are more likely to be suggested for these streams.

- **Chemistry**

Scores in chemistry had a moderate impact. The SHAP plots (in Fig. 13) reveal an interesting relationship: average scores indicate a smaller influence, whereas very high or very low scores typically have a bigger impact. This implies that performance in extreme chemistry, whether favorable or poor, serves as a differentiator when it comes to course assignment.

- **Maths**

Across several engineering specialties, mathematics has the most consistently high SHAP values out of all the attributes (as shown in Fig. 14). The importance of mathematics in terms of engineering eligibility and appropriateness is therefore reaffirmed. The recommendation probability for engineering specialties that primarily rely on mathematics, such as computer science engineering, electronics engineering, etc., is highly connected with higher math grades. Significantly higher SHAP values indicate that students who score

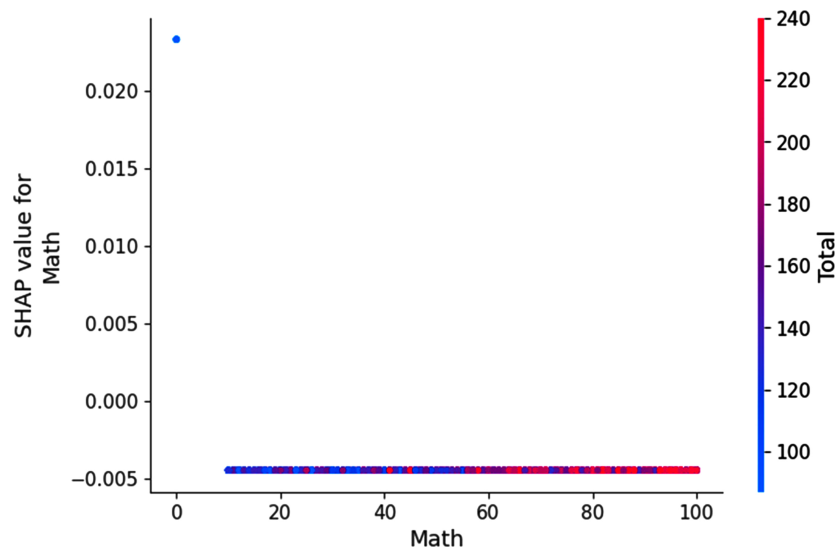


Fig. (14). SHAP dependence value for maths. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

higher than a certain threshold are more likely to be recommended for these programs.

CONCLUSION AND FUTURE SCOPE

The main objective of this study was to create a better recommendation system that would overcome the drawbacks of the conventional course recommender techniques currently in use and boost the precision and applicability of the suggestions made. Better results are achieved when the recommended models such as, KNN, Logistic Regression, Decision Trees, and Deep Learning, are applied in an ensemble manner.

In this research, we suggest Interpretability-Driven Course Recommendation: A Random Forest Approach with Explainable AI, a Course-Recommendation System for 12th-class students based on content filtering, which considers marks/grades of subjects of each student and course ratings, also for highly suitable courses to students. Not only this, but if there is any student who is not aware of the proposed system or does not have marks for subjects, then that student can express their interest. Based on this interest and the ratings of courses, the most suitable top-3 courses will be suggested to them. In this way, the cold-start problem is solved by this model. Additionally, the reason behind suggesting those courses (*i.e.*, total marks obtained by student) is also shown here. In this work, the three performance parameters precision, recall, and F1-measure are employed to measure the performance of these systems. On the basis of experimental findings, it can be said that the Random Forest Algorithm outperforms other models in the circumstances discussed, showing improvements in precision, recall, and F1-measure values.

The integration of SHAP-based explainability provides transparency, making the recommendation system more interpretable for students and educators. Additionally, since no

student can get the suggested courses of another student until he/she is having the name and mobile number of the desired student, in this way, the proposed model provides the security to the users. However, our proposed model faces some challenging issues. First, the scarcity problem is not solved by our model, and second, it is not a multilingual model. To improve its features, these two problems can be solved as future research work. Not only this, but for improving the recommendation results, this work can be expanded to create hybrid and improved models. This work can also be used to explore the analysis outcomes in a number of deep learning models.

AUTHORS' CONTRIBUTIONS

TN conceptualized the study, designed the methodology and drafted the manuscript. SZ conducted the experiments and collected data. SSB reviewed and edited it. IH supervised the project.

LIST OF ABBREVIATIONS

DCNN	= Deep Convolved Neural Network
DL	= Deep Learning
DLCRS	= Deep Learning-based Course Recommendation System
ELMO	= Embeddings from Language Models
XAI	= Explainable AI
HHO	= Harris Hawks Optimization
PRS	= Personalized Recommendation System
SHAP	= Shapley Additive explanations

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

The data and supportive information are available within the article.

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CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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